JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

IIT JAM 2008 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
15	90 M	60 Min	MCQ

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2008 • Q1 • ID: JAM_2008_Q1) MCQ +6 M Neg: −2.0

Q.1 The least positive integer n, such that $\begin{pmatrix} \cos\frac{\pi}{4} & \sin\frac{\pi}{4} \\ -\sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{pmatrix}^n$ is the identity matrix of order 2, is

(A) 4 (B) 8 (C) 12 (D) 16

Question 2 (JAM 2008 • Q2 • ID: JAM_2008_Q2) MCQ +6 M Neg: −2.0

Q.2 Let $S = \{T : \mathbb{R}^3 \to \mathbb{R}^3 \mid T$ is a linear transformation with $T(1,0,1) = (1,2,3)$, $T(1,2,3) = (1,0,1)\}$. Then S is

(A) a singleton set
(B) a finite set containing more than one element
(C) a countably infinite set
(D) an uncountable set

Question 3 (JAM 2008 • Q3 • ID: JAM_2008_Q3) MCQ +6 M Neg: −2.0

Q.3 Let $s_n = \int_0^1 \frac{n x^{n-1}}{(1+x)}\,dx$ for $n \geq 1$. Then as $n \to \infty$, the sequence $\{s_n\}$ tends to

(A) 0 (B) $1/2$ (C) 1 (D) $+\infty$

Question 4 (JAM 2008 • Q4 • ID: JAM_2008_Q4) MCQ +6 M Neg: −2.0

Q.4 The work done by the force $\vec{F} = 4y\,\hat{i} - 3xy\,\hat{j} + z^2\,\hat{k}$ in moving a particle over the circular path $x^2 + y^2 = 1,\ z = 0$ from $(1,0,0)$ to $(0,1,0)$ is

(A) $\pi + 1$ (B) $\pi - 1$ (C) $-\pi + 1$ (D) $-\pi - 1$

Question 5 (JAM 2008 • Q5 • ID: JAM_2008_Q5) MCQ +6 M Neg: −2.0

Q.5 The set of all boundary points of $\mathbb{Q}$ in $\mathbb{R}$ is

(A) $\mathbb{R}$ (B) $\mathbb{R} \setminus \mathbb{Q}$ (C) $\mathbb{Q}$ (D) $\varnothing$

Question 6 (JAM 2008 • Q6 • ID: JAM_2008_Q6) MCQ +6 M Neg: −2.0

Q.6 Let $V=\left\{(x,y,z)\in\mathbb{R}^3 : \frac{1}{4}\le x^2+y^2+z^2\le 1\right\}$ and $\vec{F}=\dfrac{x\,\hat{i}+y\,\hat{j}+z\,\hat{k}}{(x^2+y^2+z^2)^2}$ for $(x,y,z)\in V$. Let $\hat{n}$ denote the outward unit normal vector to the boundary of V and S denote the part $\left\{(x,y,z)\in\mathbb{R}^3 : x^2+y^2+z^2=\frac{1}{4}\right\}$ of the boundary of V. Then $\iint_S \vec{F}\cdot\hat{n}\, dS$ is equal to

(A) -8π (B) -4π (C) 4π (D) 8π

Question 7 (JAM 2008 • Q7 • ID: JAM_2008_Q7) MCQ +6 M Neg: −2.0

Q.7 The set $U=\left\{x\in\mathbb{R} \mid \sin x=\frac{1}{2}\right\}$ is

(A) open (B) closed
(C) both open and closed (D) neither open nor closed

Question 8 (JAM 2008 • Q8 • ID: JAM_2008_Q8) MCQ +6 M Neg: −2.0

Q.8 Let $f(x)=\int_0^x (x^2-t^2)\,g(t)\,dt$, where g is a real valued continuous function on $\mathbb{R}$. Then $f'(x)$ is equal to

(A) 0 (B) $x^3\,g(x)$ (C) $\int_0^x g(t)\,dt$ (D) $2x\int_0^x g(t)\,dt$

Question 9 (JAM 2008 • Q9 • ID: JAM_2008_Q9) MCQ +6 M Neg: −2.0

Q.9 Let $y_1(x)$ and $y_2(x)$ be linearly independent solutions of the differential equation $y''+P(x)y'+Q(x)y=0$, where $P(x)$ and $Q(x)$ are continuous functions on an interval I. Then $y_3(x)=a\,y_1(x)+b\,y_2(x)$ and $y_4(x)=c\,y_1(x)+d\,y_2(x)$ are linearly independent solutions of the given differential equation if

(A) $ad=bc$ (B) $ac=bd$ (C) $ad\neq bc$ (D) $ac\neq bd$

Question 10 (JAM 2008 • Q10 • ID: JAM_2008_Q10) MCQ +6 M Neg: −2.0

Q.10 The set $R=\{f \mid f \text{ is a function from } \mathbb{Z} \text{ to } \mathbb{R}\}$ under the binary operations $+$ and $\cdot$ defined as $(f+g)(n)=f(n)+g(n)$ and $(f\cdot g)(n)=f(n)\,g(n)$ for all $n\in\mathbb{Z}$ forms a ring. Let $S_1=\{f\in R \mid f(-n)=f(n) \text{ for all } n\in\mathbb{Z}\}$ and $S_2=\{f\in R \mid f(0)=0\}$. Then

(A) S_1 and S_2 are both ideals in R

(B) S_1 is an ideal in R while S_2 is not

(C) S_2 is an ideal in R while S_1 is not

(D) neither S_1 nor S_2 is an ideal in R

Question 11 (JAM 2008 • Q11 • ID: JAM_2008_Q11) MCQ +6 M Neg: −2.0

Q.11 Let $T:\mathbb{R}^3\to\mathbb{R}^3$ be a linear transformation such that $T(1,2,3)=(1,2,3)$, $T(1,5,0)=(2,10,0)$ and $T(-1,2,-1)=(-3,6,-3)$. The dimension of the vector space spanned by all the eigenvectors of T is

(A) 0 (B) 1 (C) 2 (D) 3

Question 12 (JAM 2008 • Q12 • ID: JAM_2008_Q12) MCQ +6 M Neg: −2.0

Q.12 Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers defined as $a_1=1$ and for $n\geq 1$, $a_{n+1}=a_n+(-1)^n 2^{-n}$, $b_n=\dfrac{2a_{n+1}-a_n}{a_n}$. Then

(A) $\{a_n\}$ converges to zero and $\{b_n\}$ is a Cauchy sequence

(B) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is a Cauchy sequence

(C) $\{a_n\}$ converges to zero and $\{b_n\}$ is not a convergent sequence

(D) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is not a convergent sequence

Question 13 (JAM 2008 • Q13 • ID: JAM_2008_Q13) MCQ +6 M Neg: −2.0

Q.13 Let $f:(-1,1)\to\mathbb{R}$ be defined as $f(x)=\dfrac{x^2}{1-\cos x}$ for $x\neq 0$ and $f(0)=2$. If $f(x)=\sum_{n=0}^{\infty}a_n x^n$ is the Taylor expansion of f for all x in $(-1,1)$, then $\sum_{n=0}^{\infty}a_{2n+1}$ is

(A) 0 (B) $1/2$ (C) 1 (D) 2

Question 14 (JAM 2008 • Q14 • ID: JAM_2008_Q14) MCQ +6 M Neg: −2.0

Q.14 Let $y_1(x)$ and $y_2(x)$ be twice differentiable functions on an interval I satisfying the differential equations $\frac{dy_1}{dx} - y_1 - y_2 = e^x$ and $2\frac{dy_1}{dx} + \frac{dy_2}{dx} - 6y_1 = 0$. Then $y_1(x)$ is

(A) $C_1 e^{-2x} + C_2 e^{3x} - \frac{1}{4}e^x$

(B) $C_1 e^{2x} + C_2 e^{-3x} + \frac{1}{4}e^x$

(C) $C_1 e^{2x} + C_2 e^{-3x} - \frac{1}{4}e^x$

(D) $C_1 e^{-2x} + C_2 e^{3x} + \frac{1}{4}e^x$

Question 15 (JAM 2008 • Q15 • ID: JAM_2008_Q15) MCQ +6 M Neg: −2.0

Q.15 Let G be a finite group and H be a normal subgroup of G of order 2. Then the order of the center of G is

(A) 0 (B) 1 (C) an even integer ≥ 2 (D) an odd integer ≥ 3

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2008 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	**B**
2	MCQ	+6	**D**
3	MCQ	+6	**B**
4	MCQ	+6	**D**
5	MCQ	+6	**A**
6	MCQ	+6	**A**
7	MCQ	+6	**B**
8	MCQ	+6	**D**

Q. No.	Type	Marks	Official Key
9	MCQ	+6	**C**
10	MCQ	+6	**C**
11	MCQ	+6	**D**
12	MCQ	+6	**B**
13	MCQ	+6	**A**
14	MCQ	+6	**C**
15	MCQ	+6	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.